



SHAHEED ZULFIKAR ALI BHUTTO

INSTITUTE OF SCIENCE AND TECHNOLOGY

Software Requirements Specification (SRS)

TravelGuide

Tour Planning Android Application

Version 1.0 Approved

Students: Munir Ahmed (2212138) & Meer Jaan (2212133)

Project Supervisor: Mr. Sufyan Haqqi

Organization: Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

Date: 16 March, 2026

Table of Contents

Revision History	ii
1. Introduction.....	1
1.1 Purpose.....	1
1.2 Document Conventions.....	1
1.3 Intended Audience and Reading Suggestions.....	1
1.4 Product Scope	1
1.5 References.....	1
2. Overall Description.....	2
2.1 Product Perspective.....	2
2.2 Product Functions	2
2.3 User Classes and Characteristics.....	2
2.4 Operating Environment.....	2
2.5 Design and Implementation Constraints	2
2.6 User Documentation	2
2.7 Assumptions and Dependencies.....	2
3. External Interface Requirements.....	3
3.1 User Interfaces	3
3.2 Hardware Interfaces	3
3.3 Software Interfaces	3
3.4 Communications Interfaces.....	3
4. System Features	3
4.1 User Account Management.....	3
4.2 Hotel Module	4
4.3 Restaurant Module	4
4.4 Transport Module.....	5
4.5 Budget Estimation Module	5
4.6 Admin Module	6
5. Other Nonfunctional Requirements	6
5.1 Performance Requirements	6
5.2 Safety Requirements	6
5.3 Security Requirements.....	6
5.4 Software Quality Attributes	7
5.5 Business Rules	7
6. Other Requirements	7
Appendix A: Glossary.....	7
Appendix B: Analysis Models.....	7
Appendix C: To Be Determined List	7

Revision History

Name	Date	Reason for Changes	Version
Munir Ahmed & Meer Jaan	2026	Initial Draft	1.0

1. Introduction

1.1 Purpose

This SRS presents the general non-functional and functional requirements of the TravelGuide mobile application version 1.0 that is written in Android. It discusses the integrated Travelers, Hotel Owners, Restaurant Owners, Transport Providers and Admin modules of an Android mobile environment.

1.2 Document Conventions

All requirements are assigned unique IDs: REQ-XX is a functional requirement and NFR-XX is non-functional requirement. The levels of priority are High, Medium and Low. Application or app are used in this document to mean the TravelGuide Android mobile application.

1.3 Intended Audience and Reading Suggestions

The report is written to the FYP development team, project supervisor and evaluators. Developers will read all the sections. These are the Sections 1, 2 and 4 that should be paid attention to by supervisors and evaluators. Sections 4 and 5 should be given attention by testers.

1.4 Product Scope

The Android phone application, TravelGuide is the solution that will aid in resolving the problem of scattered travel information within Pakistan. It provides a single mobile interface where the travelers might compare the prices of the hotels, food prices and travel fares. The application will automatically calculate the approximate cost of traveling so that it becomes easy to plan a trip all the way with an Android smart phone.

1.5 References

- [1] S. Sharma and A. Gupta, "Design and development of a mobile-based travel planning system," IJCSIT, vol. 9, no. 3, pp. 45–49, 2018.
- [2] M. Patel and R. Shah, "Mobile travel management systems for budget estimation," IJACSA, vol. 10, no. 5, pp. 210–215, 2019.
- [3] IEEE Std 830-1998, "IEEE Recommended Practice for Software Requirements Specifications," IEEE, 1998.

2. Overall Description

2.1 Product Perspective

TravelGuide is a free Android app. It is developed using a React Native front-end (Android), Node.js/Express.js backend REST API and MySQL database and firebase Firestore real-time synchronization. The users can access the system on the native android application in their smartphones. The backend is linked to the app with HTTPS.

2.2 Product Functions

- Registration and login of the Travelers, Service Providers and Admin under role-based access control.
- Service providers upload, update and delete their uploads and description of price to the mobile app.
- Using Android application, travelers read and find various services offered by the hotels, restaurants and transport services by location.
- Customers will select services and design an end-to-end trip experience through their Android mobile.
- Android application will automatically determine and show the overall estimated travel budget in real-time.
- Travelers save their trip plans, which can be accessed at any time later and can be made available when there is no connection.
- Android device location GPS discovery of nearby services.
- Admin manages all the user accounts and service listing via a central mobile dashboard.

2.3 Classes and Characteristics of users.

Android application consists of five classes of users:

Travelers: Typical users who sign up to search services, create trip plans, and see budget estimates on their Android smartphones.

Hotel Owners: Sign up as service providers to add hotel properties and room categories and prices.

Restaurant Owners: Sign up to add restaurants by cuisine type and average meal price.

Transport Providers: Register to add transport services including routes and fares.

Admin: Can administer the system with full access to manage all users and listings via the Android Admin Panel.

2.4 Operating Environment

Client Side: Android devices with Android 9.0 (API level 28) or higher.

Server Side: Linux server with Node.js 18.x and MySQL 8.0+.

Technical Stack: React Native (Android), Node.js/Express.js server, MySQL database, Firebase. Firestore, JWT authentication.

2.5 Constraints of the design and implementation.

- The mobile app should be developed in React Native with Android (API level 28+) target.
- MySQL should be taken as the relational database. All authentication has to be done by JWT.
- The JWT token should be saved in AsyncStorage of the Android phone.
- Traveler, Provider and Admin functionality should be clearly divided in role-based access control.
- The user passwords should be stored in the form of an bcrypt hash. Storage in plain text is not allowed.
- The app should be able to have simple offline support with AsyncStorage and Firebase Firestore offline support.

2.6 User Documentation

The documentation will include the following: a User Manual guide to travelers to explain how to use the Android application., a Provider Guide to the service providers on how to maintain listings as well as the Admin Guide to the system administrator. Key screens will also have in-app help tooltips.

2.7 Assumptions and Dependencies

- An Android smartphone with a stable mobile data or Wi-Fi connection to synchronize data is available to all users and on Android 9.0 or later.
- Service providers have the sole responsibility of the correctness of their price, descriptions, and availability.
- The budget produced is a rough estimate of the budget used to plan and is not a definite booking and a price guarantee.
- The Node.js API and MySQL database are continuously being availed in the background to accept requests made by the Android client.

3. External Interface Requirements

3.1 User Interfaces

- There has to be a Bottom Navigation Bar on all the main screens that has links to: Home, Hotels, Restaurants, Transport and Profile.
- The trip plan screen should display all the chosen services and automatically generated budget summary to view on Android mobile devices.
- All screens should have standard Android back navigation and should obey the Android gesture navigation bar.

3.2 Hardware Interfaces

The Android application needs a smartphone with at least 3 GB RAM, a GPS device to locate the services and a minimum resolution of 1080×1920 on a screen. The application makes use of the Android GPS hardware via the React Native Maps library.

3.3 Software Interfaces

The React Native Android client is connected to the Node.js/Express.js backend through RESTful HTTPS API calls. Firebase Firestore SDK is a React Native-based module to synchronize data in real-time. AsyncStorage offers local offline storage.

3.4 Communications Interfaces

- The interactions between server and Android app are based on REST API and HTTPS in production and HTTP in local development.
- The JWT authentication scheme is the Bearer token scheme that is stored in the Android device in the AsyncStorage.

4. System Features

4.1 User Account Management

4.1.1 Description and Priority

This aspect includes user registration, log in, profile management and log out in the Android application. It forms the backbone of the whole application as all the other features rely on the authenticated users whose role is assigned correctly.

Priority: High

4.1.2 Stimulus / Response Sequences

1. User completes the registration form on the Android application and sends it → system checks the inputs and establishes the account.
2. User logs in to the application by providing email and password on the front screen of the application - the system checks the credentials and sends a JWT token, which is stored in AsyncStorage.
3. User taps logout, system removes the JWT token in the AsyncStorage and takes them to the login screen.

4.1.3 Functional Requirements

Req ID	Description	Priority	Module	Status
REQ-01	Registration is possible by any user by giving name, email, password and role (Traveler or Service Provider). Email should be unique.	High	Auth	Approved
REQ-02	Email and password can be used to log in by registered users. An AsyncStorage on the Android device is created and houses a JWT token.	High	Auth	Approved
REQ-03	The system only allows access by user role. Every role is only allowed to access their authorized screens.	High	Auth	Approved
REQ-04	The Profile screen allows the user to update his/her profile information such as name, contact number and password.	Medium	Auth	Approved
REQ-05	Users can log out. The JWT token will be removed out of AsyncStorage when one logs out.	High	Auth	Approved

4.2 Hotel Module

4.2.1 Description and Priority

Using this feature, the hotel owners can control the hotel listing and travelers to locate hotels using trip plans using the Android application.

Priority: High

4.2.2 Stimulus / Response Sequences

1. Fill in the Add Hotel form on mobile app → submit → save listing.
2. The traveler clicks into the destination and clicks search the system will give a list of matched hotels with details and prices.
3. Traveler selects a hotel and type of number of nights at hotel into system, causes hotel cost to be added to trip plan.

4.2.3 Functional Requirements

Req ID	Description	Priority	Module	Status
REQ-06	It is possible to create a listing with hotel name, location, description, type of rooms and price per night via the mobile application.	High	Hotel	Approved
REQ-07	The owners of the hotels have an option to update the information and cost of their existing hotel listings.	High	Hotel	Approved
REQ-08	Hotel owners have the ability to get rid of their listings in the platform permanently.	Medium	Hotel	Approved
REQ-09	Users have the opportunity to search through hotels available and filter the results according to their location. Hotel pins are shown on the map screen.	High	Hotel	Approved
REQ-10	Users are able to choose a hotel and indicate the number of nights to include the hotel in their trip planner.	High	Hotel	Approved

4.3 Restaurant Module

4.3.1 Description and Priority

This will enable restaurant owners to update and control the listing and enable travelers to search and choose restaurants to plan their trips.

Priority: High

4.3.2 Stimulus / Response Sequences

1. The owner of the restaurant completes the Add Restaurant form and sends it to the system → listing is saved in the system.
2. Traveler searches by destination → system presents suitable restaurants with menu information and prices.

3. Traveler chooses a restaurant and the number of meals approximate to consume during the trip- the system includes the cost of food to a trip plan.

4.3.3 Functional Requirements

Req ID	Description	Priority	Module	Status
REQ-11	The restaurant owners could add a listing of their names, location, type of cuisine, menu items and price per meal.	High	Rest.	Approved
REQ-12	Restaurant owners have an opportunity to modify information about their restaurants and change menu prices.	High	Rest.	Approved
REQ-13	The owners of restaurants are able to remove their restaurant listing on the site.	Medium	Rest.	Approved
REQ-14	On the Android application, travelers can browse restaurants and narrow down to any location or type of cuisine.	High	Rest.	Approved
REQ-15	A traveler is able to choose a restaurant and input estimated food to make it a part of his trip plan.	High	Rest.	Approved

4.4 Transport Module

4.4.1 Description and Priority

This option enables transport operators to control the listings and the travelers to view and choose transport services.

Priority: High

4.4.2 Stimulus / Response Sequences

1. Add transport form is filled in by the transport provider and listing saved with vehicle type, route, and fare.
2. Search by route or vehicle type by the traveler → system provides a corresponding transport service.

3. Traveler picks a transport service → system charges the cost of transport to the trip plan.

4.4.3 Functional Requirements

Req ID	Description	Priority	Module	Status
REQ-16	Providers of transport can include a listing of the type of vehicle, route, departure time, and fare per head.	High	Trans.	Approved
REQ-17	Transport providers are able to update their transport listings and change fares.	High	Trans.	Approved
REQ-18	Transport providers will have the ability to remove their transport listings.	Medium	Trans.	Approved
REQ-19	With the Android application, travelers have the option of browsing transport services and filtering by route or vehicle type.	High	Trans.	Approved
REQ-20	Customers have an option of a transport service to include in their trip plan.	High	Trans.	Approved

4.5 Budget Estimation Module

4.5.1 Description and Priority

This is the distinguishing characteristic of the Android app. It automatically calculates the sum of the approximate travel budget each time a traveler chooses or changes a service in his or her trip arrangement.

Priority: High

4.5.2 Stimulus / Response Sequences

1. Traveler adds a hotel to the trip plan, the app automatically changes the budget summary.
2. Visitor includes a restaurant or transport, app re-calculates and shows the new total.
3. Traveler deletes a service → app recalculates and changes the display of the budget in real time.
4. Traveler taps Save Trip → the entire trip plan and budget estimate is saved in the database.

4.5.3 Functional Requirements

Req ID	Description	Priority	Module	Status
REQ-21	Android application can automatically compute the total cost of a trip by adding hotel, restaurant and transport cost.	High	Budget	Approved
REQ-22	A breakdown of the budget is presented in items showing every service and its price and the total amount that is estimated.	High	Budget	Approved
REQ-23	The budget estimate is automatically updated each time a traveler adds or removes a service.	High	Budget	Approved
REQ-24	The traveler is able to store his entire trip plan and budget estimate as a reference.	Medium	Budget	Approved
REQ-25	Users can see the trip plans they had previously saved on the Profile screen even in offline mode.	Medium	Budget	Approved

4.6 Admin Module

4.6.1 Description and Priority

The feature offers the administrator with the means to control all the user accounts and service listings via the Android Admin Panel.

Priority: Medium.

4.6.2 Stimulus / Response Sequences

1. Admin logs in → redirected to Admin Panel with a total user count, listings, and saved trip plans.
2. Admin locates a user account- looks at the details of the account and deactivates or deletes the account.
3. Admin discovers a listing that breaks the rules of the platform and deletes it.

4.6.3 Functional Requirements

Req ID	Description	Priority	Module	Status
REQ-26	Admin will access a dashboard summary of total registered users, total active listings and total saved trip plans.	Medium	Admin	Approved
REQ-27	The Data can be managed by the administrator, who can see all the user accounts registered and can deactivate, activate or permanently delete any account.	High	Admin	Approved
REQ-28	Admin is able to see all services listings by all the providers and can edit or delete any listing	High	Admin	Approved
REQ-29	Budget limit alerts are sent out as push notifications through Android notification system.	High	Admin	Approved
REQ-30	The app takes advantage of the Android device GPS to propose the closest hotels and restaurants depending on the current position of the traveler.	Medium	Admin	Approved
REQ-31	Cached data allow travelers to see their trip plans and budget summaries without being online.	High	Admin	Approved

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Description	Metric / Target	Priority
NFR-01	All screens in the Android application should be loaded in 3 seconds in a typical 4G/Wi-Fi connection.	Screen load $\leq$ 3 sec	High
NFR-02	The calculation of the budget should be done, and the results should be shown within 1 second of the traveler choosing or eliminating a service.	Response $\leq$ 1 sec	High
NFR-03	The system should have the capability to handle at least 50 concurrent users without a significant drop in performance.	50 concurrent users	Medium

5.2 Safety Requirements

Android application is not associated with any physical hardware and does not have any serious safety issues. Nevertheless: user personal data should not be made accessible to any unauthorized user; any error messages should not reveal anything about the inner workings of the system or about the database content; and all user-supplied data should be sanitized prior to processing in order to avoid injection attacks.

5.3 Security Requirements

ID	Description	Metric / Target	Priority
NFR-04	bcrypt hashing should be used to store all user passwords. Storage of plain text passwords is not allowed.	bcrypt required	High
NFR-05	Any secured API routes should have a valid JWT token. Without a legitimate token, 401 should be sent.	JWT on all routes	High

NFR-06	Role-based access control should not allow travelers to have access to provider or admin screens, and vice versa.	RBAC enforced	High
NFR-07	All user input should be validated and sanitized both in Android front and backend to avoid SQL injection and XSS.	Input validation	High

5.4 Software Quality Attributes

ID	Description	Metric / Target	Priority
NFR-08	Usability: A new user has to be able to sign up and plan a trip on the Android app without any outside instructions.	No training required	High
NFR-09	Reliability: The budget calculation module should never give wrong results with a zero error in calculation.	Zero calculation errors	High
NFR-10	Maintainability: The codebase must be organized into separate modules (Hotel, Restaurant, Transport, Auth, Budget).	Modular architecture	Medium
NFR-11	Availability: The Android application backend should be available during 99% of any calendar month.	99% uptime	High
NFR-12	Compatibility: The app should be compatible with the standard android screen resolution of 5-inch to 6.7-inch displays.	Android compatibility	Medium
NFR-13	Efficiency of the battery: The app should be optimized to consume minimum background battery when the services of GPS are in use.	Minimal battery drain	Medium

5.5 Business Rules

1. Android application is accessible to only registered users. Browsing is not allowed as a guest.
2. One user can be in a single role (Service Provider or Traveler). Admin approval is needed in terms of role changes.
3. Service providers have access only to their own listings and not that of other providers.
4. The budget estimate is obviously stated to be approximate and not a binding booking or payment.
5. Admin can see all platform data, but is not able to change the logic of budget calculations.

6. Other Requirements

1. **Database:** In MySQL, there are all the data with appropriate foreign key association between Users, Listings, Trip Plans, and Budgets.
2. **Language:** Version 1.0 is only English. In a future version Urdu language support can be added.
3. **Error Logging:** The backend should record all the major errors in a log file to debug and maintain.
4. **Data Backup:** MySQL database should also be backed up at a regular frequency to avoid the loss of data.

Appendix A: Glossary

This document is called SRS - Software Requirements Specification.

- FYP Final Year Project.
- API -Application Programming Interface.
- JWT — JSON Web Token, a secure authentication that is stored in the Android device in the AsyncStorage.
- RBAC Role-Based Access Control, limits accessibility of features according to user role.
- AsyncStorage - Local storage on the Android device of JWT and cached data.
- React Native - JavaScript platform to create native Android (and iOS) applications.
- Traveler - A registered user that uses the Android app to plan travel.

- Service Provider- A hotel owner, restaurant owner or transport operator registered on the platform.
- Plan of trip - Services chosen by a traveler and the approximate budget.
- Budget Estimate Budget estimate is the estimated total cost of a planned trip, automatically calculated.

Appendix B: Analysis Models

1. **Entity-Relationship:** Main entities are User, Hotel, Restaurant, Transport, TripPlan, and Budget. A TripPlan belongs to one Traveler and references selected services. A Budget record is linked to one TripPlan.
2. **Data Flow:** Traveler selects services on Android app → App sends data to Backend API → Backend retrieves pricing from MySQL → Budget Engine calculates total → Result displayed on Android screen.
3. **Use Case Summary:** Register, Login, Add/Edit/Delete Listing, Search Services, Build Trip Plan, View Budget Estimate, Save Trip Plan (offline accessible), Manage Users (Admin), Manage Listings (Admin).

Appendix C: To Be Determined List

1. **TBD-01:** Whether an online payment gateway will be integrated for confirmed bookings in a future version.
2. **TBD-02:** The specific push notification service to be used for Android budget alert notifications.
3. **TBD-03:** Whether the app will be published to Google Play Store in a future version.
4. **TBD-04:** The specific cloud hosting provider for production deployment of the backend.